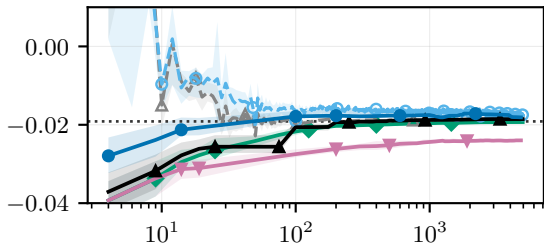
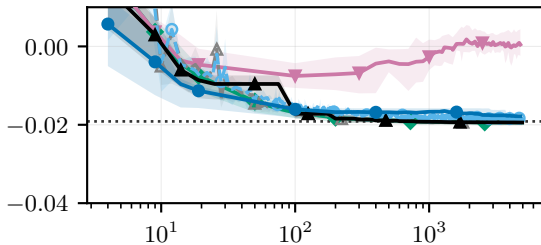
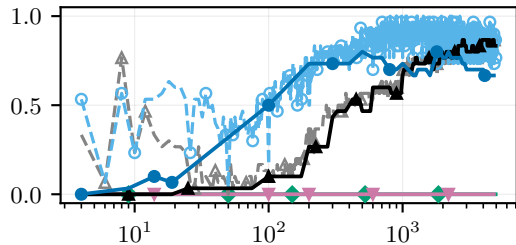


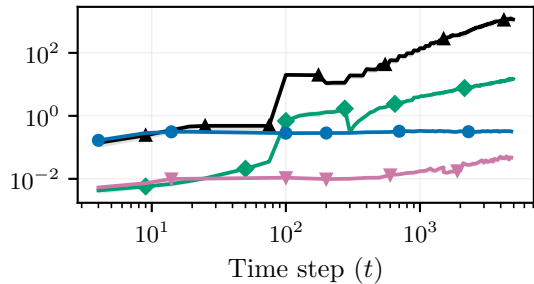
In-sample objective value



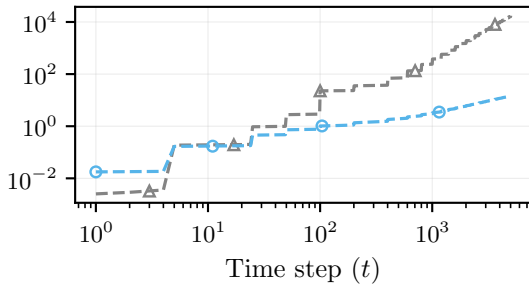
Out-of-sample objective value

Confidence  $1 - \hat{\beta}_t$ 

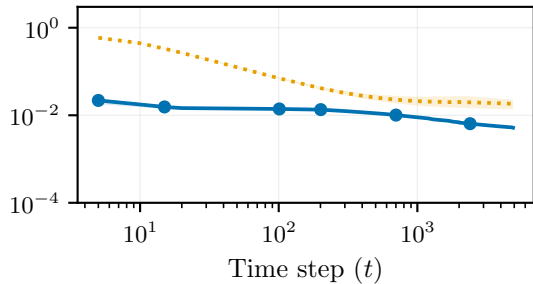
Per-iteration time (s), exact



Cumulative time (s), online



Online clustering dynamic regret



—●— online clustering

—▲— DRO

—◆— SAA

.....  $H^*$ 

- - - upper bound

- - -○- online clustering subgrad

- - -▲- DRO subgrad

- - -▼- cluster SAA

—●— empirical regret